

5 Finding impulse response

Note: From now on, I'll write convolution with $*$.

Say we have a system

$$y[n] = b_0x[n] + b_1x[n-1] - ay[n-1]$$

This system is causal, only depending on previous values.

$h[n]$, the impulse response, is $y[n]$ when $x[n] = \delta[n]$.

We have initial conditions, where for $n < 0$, $h[n] = 0$.

$$h[n] = -a_1h[n-1] + b_0\delta[n] + b_1\delta[n-1]$$

We can solve this iteratively and make a table

n	h[n]
<0	0
0	b_0
1	$b_0 - a_1b_0$
2	$a_1b_1 - a^2b_0$
...	

The pattern for this is

$$h[n] = \begin{cases} 0 & n < 0 \\ b_0 & n = 0 \\ (a_1)^{n-1}b_1 + (-a_1)^nb_0 & n \geq 1 \end{cases}$$

Which we can write with step functions

$$h[n] = b_0\delta[n] + ((-a_1)^{n-1}b_1 + (-a_1)^nb_0)u[n]$$

There is a better way to do this though.

We can consider periodic functions $x(t)$

If we can write it as a Fourier series, like:

$$x(t) = x(t + T_0) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \text{ where } \omega_0 = \frac{2\pi}{T_0}$$

I.e., $y(t) = \cos(\omega t) * h(t)$

$$\frac{1}{2}(e^{j\omega t} + e^{-j\omega t})$$

$$H(j\omega) = \frac{1}{1+j\omega}$$

So we get

$$H(j\omega) \frac{1}{2} e^{j\omega t} + \frac{H(-j\omega)1}{2} e^{-j\omega t}$$

We take the continuous time Fourier series, where

$$a_k = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) e^{-jk\omega_0 t} dt$$